INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

CLOTH, FLANNEL, WOOL AND COTTON, WOOL AND NYLON

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-C-3760.

Abstract. This commercial item description covers flannel cloth intended for use as lining material in various military clothing items. The type III cloth is intended for use as lining material in Cap, Combat, Woodland, (MIL-C-1911).

1. CLASSIFICATION

1.1 Classification. The cloth shall be of the following types as specified (see 5.1).

Type I - 12-ounce, wool and cotton, plain weave
Type II - 20-ounce, wool and cotton, twill weave
Type III - 12-ounce, wool and nylon, plain weave

2. SALIENT CHARACTERISTICS

2.1 General description. The yarns, for the types I and II cloth, shall be made from a blend of not less than 80 percent wool (not lower in grade than 60's U.S. Standard) and not more than 20 percent cotton. The yarns, for the type III cloth shall be made from a blend of not less than 80 percent wool (not lower in grade than 60's U.S. Standard) and not more than 20 percent staple nylon. The wool portion of the types I, II, and III yarns shall be such that the finished yarn shall contain:

not less than 60 percent fleece or pulled wool, or any blend of both,
not more than 10 percent wool noils, and
not more than 10 percent reprocessed, reused, or reworked wool.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATIN: STRNC-IRT, Natick, MA 01760-5019.

AMSC N/A FSC 8305

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The width shall be 60 inches, minimum, exclusive of the selvage. The cloth shall be scoured, fulled, napped, and sheared on the back and the type and character of finish shall conform to that shown by the standard sample. The color of the finished cloth shall be as specified (see 5.1) and shall be obtained by stock or piece dyeing. The cloth shall be mothproofed in accordance with MIL-C-43665 during the dyeing operation. The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

2.2 Physical requirements. The finished cloth shall conform to the requirements listed in table I.

TABLE I. Physical requirements

Characteristic	Requirements Type I	Type II	Type III	Test method 1/
Fiber content percent:				
Wool (min)	80	80	80	-
Cotton (max)	20	20	-	-
Nylon (max)	-	-	20	-
Colorfastness to:				
Light (after 40 standard fading hours)	Good 2/	Good 2/	Good 2/	5660
Perspiration	Good 2/	Good 2/	Good 2/	5680
Wet drycleaning	Good 2/	Good 2/	Good 2/	5622
Crocking	3.5 3/	3.5 3/	-	5651
Labile sulfur content	4/	4/	-	2020
Weave	Plain	2/2 right hand twill	Plain	Visual
Weight, oz. (min) 5/	12.0	20.0	12.0	5041
Yarns per inch (min):				
Warp	38	46	38	5050
Filling	31	44	31	5050
Breaking strength, lbs. (min):				
Warp	42	60	42	5100
Filling	32	50	32	5100

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TABLE I. Physical requirements (cont'd)

Characteristic	Requirements Type I	Type II	Type III	Test method 1/
pH	6/	6/	6/	2811
Shrinkage, percent (max):				
Warp	6.0	6.0	6.0	5590
Filling	4.0	4.0	4.0	5590

1/ The test methods indicated, from FED-STD-191 Textile Test Methods, have been found acceptable to determine if the cloth meets the stated requirements. If the government requires proof that the cloth meets the stated requirements, the test methods indicated shall be used.

2/ Or equal to or better than the standard sample.

3/ Refers to AATCC Chromatic Transference Scale rating.

4/ The dyed and finished cloth shall contain no more labile sulfur than shown by standard sample. When a standard sample is not available the cloth shall show not more than a "slight" trace of labile sulfur as defined in the test method.

5/ Based on linear yard of 56-inch width cloth.

6/ The pH value of the water extract of the finished cloth shall be not less than 4.0 and not more than 8.0.

2.3 Length and put-up. Unless otherwise specified (see 5.1), the cloth shall be furnished in continuous lengths, each not less than 50 yards for types I and III, and not less than 40 yards for type II. The cloth shall be put-up full width on rolls as specified in 4.1. Each roll of cloth shall be stamped "back" on the back side of the cloth at the end of each roll.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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3.2 Visual examination. Lots of cloth shall be inspected in accordance with MIL-STD-105. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5. The lot size shall be expressed in units of yards. The sample unit shall be one linear yard. The cloth shall be examined for the following defects: hole, cut, tear, smash, multiple floats, broken or missing end or pick, embedded crease or wrinkle, spots or stains.

3.3 Length examination. During the visual examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the roll ticket shall be considered a length defect. If two or more rolls in the sample are defective with respect to length or if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the roll tickets, the lot shall be rejected.

3.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Packaging. The cloth shall be rolled on a convolute or spiral-wound chipboard tube. The tube shall have a smooth finish surface that will prevent the fibers of the chipboard from transferring to the cloth. Each roll of cloth, put-up as specified, shall be wrapped with 60-pound minimum basis weight kraft paper in such a way that the paper shall encircle the roll at least once with a minimum overlap of 3 inches, and the width of the paper shall be sufficient to fold over and protect the ends of the roll. Gummed paper tape, with a minimum width of 2-1/2 inches, shall be applied to the overlap seam the full length of the roll and across each end. Alternately, each roll of cloth shall be enclosed in a polyethylene film tube, and the tube secured by heat-sealing or mechanical ties.

4.2 Packing. Each wrapped roll shall be enclosed in a bias-sewn tubing made from woven polypropylene cloth. The tube shall be secured by heat-sealing, by means of a mechanical tie, or by plastic ties with a lock-end. Alternatively, a maximum of 325 pounds of rolls of kraft-wrapped cloth shall be packed in a snug-fitting double-sided fiberboard container known as a double cover, and shall consist of a body or joined liner and two covers. Corrugated fiberboard with a minimum bursting strength of 500 pounds shall be used. The container shall be strapped with metallic straps not less than 1/2 by 0.020 inch or non-metallic straps not less than 7/16 by 0.030 inch.

4.3 Marking. In addition to any special markings required by the contract or purchase order, rolls and shipping containers shall be marked in accordance with MIL-STD-129. Each roll shall have a piece ticket (identification tag) affixed with a 5-ply minimum cotton string or a plastic tag hanger. The tag shall have a reinforced eyelet and shall be legibly printed, using water insoluble ink, with the following information:

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Procurement Agency	Contract or Purchase Order Number
National Stock Number	Finish or Greige Mill
Item Description	Roll Number
Contractor	Lot Number
Width	Piece Number
Fiber Content	Yardage

5. NOTES

5.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of cloth required (see 1.1).
c. Color of cloth required (see 2.1).
d. Length and put-up if other than specified (see 2.3).

5.2 Sources of documents.

5.2.1 Sources of nongovernment association documents.

AATCC Chromatic Transference Scale is available from:

American Association of Textile Chemists and Colorists (AATCC)
P.O. Box 12215
Research Triangle Park, NC 27709-2215

US Standard for Determining Grade of Wool Top is available from:

US Department of Agriculture
Agricultural Marketing Service
Washington, DC 20402

5.2.2 Source of Government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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MILITARY INTERESTS:

Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Air Force - 82, 99
DLA - CT

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

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